

Q1	A	B	C	D	E	F
Q2	A	B	C	D	E	F
Q3	A	B	C	D	E	F
Q4	A	B	C	D	E	F
Q5	A	B	C	D	E	F
Q6	A	B	C	D	E	F
Q7	A	B	C	D	E	F
Q8	A	B	C	D	E	F
Q9	A	B	C	D	E	F
Q10	A	B	C	D	E	F
Q11	A	B	C	D	E	F
Q12	A	B	C	D	E	F
Q13	A	B	C	D	E	F
Q14	A	B	C	D	E	F

1. Which statement best defines a "polyatomic ion"? [1]

A metal atom that is part of an acid.

A group of covalently bonded atoms that has an overall net electrical charge.

An ion formed from a single atom that has multiple possible charges.

An element that can exist in many different forms.

2. What is the correct chemical name for the simple ionic compound CaO? [1]

Calcium oxide

Calcium oxygen

Monocalcium monoxide

Calcium(II) oxide

3. What is the correct chemical name for the acid H_2SO_4 , which contains the sulfate ion (SO_4^{2-})? [1]

Sulfuric acid

Sulfurous acid

Hydrosulfuric acid

Hydrogen sulfate acid

4. The polyatomic ions nitrate (NO_3^-) and nitrite (NO_2^-) are important in naming certain acids. Both of these ions contain which element in addition to oxygen? [1]

Nitrogen

Carbon

Chlorine

Sulfur

5. In the chemical formulas of most acids, which element is typically listed first? [1]

Hydrogen

Nitrogen

Oxygen

Carbon

6. A compound formed between a metal element and a nonmetal element is generally classified as: [1]

an ionic compound

an acid

a covalent compound

single nonmetal element, what suffix is typically used for the nonmetal's name?
[1]

an element

-ite

7. What is the correct chemical name for the binary acid HBr? [1]

-ic

Hydrobromous acid

-ide

Bromic acid

-ate

Hydrogen bromide acid

Hydrobromic acid

11. What does the term "binary ionic compound" mean? [1]

8. When naming a binary acid, what prefix is typically added to the beginning of the name? [1]

An ionic compound containing two atoms of the same metal.

per-

An ionic compound containing two different polyatomic ions.

poly-

An ionic compound made of atoms from only two different elements, one metal and one nonmetal.

hydro-

An ionic compound that is twice as strong as other ionic compounds.

non-

9. If a polyatomic acid contains a polyatomic ion whose name ends in "-ate" (e.g., nitrate), what will the ending of the acid's name be? [1]

12. Which of the following chemical formulas most likely represents an acid?
[1]

hydro- -ic acid

NaCl

-ide acid

CO₂

-ic acid

H₂O

-ous acid

HCl

10. When naming a simple ionic compound consisting of a metal and a

13. Roman numerals (such as I, II, III) are used in the names of some ionic compounds to indicate: [1]

The positive charge of the metal cation,
especially when the metal can form ions
with different charges.

The negative charge of the anion.

The total number of atoms in one
molecule of the compound.

The number of nonmetal atoms in the
formula.

14. Prefixes such as 'mono-', 'di-', and 'tri-'
are commonly used in the systematic
naming of which type of compounds? [1]

Polyatomic acids (Oxyacids)

Ionic compounds containing polyatomic
ions

Binary acids

Binary covalent compounds